

# Caelaris Linux

Next-Generation High-Performance Arch-Based Linux Distribution with Dual Desktop Sessions, Custom Graphical Installer, Universal Driver Automation, and Pro-Grade Gaming Stack

|                     |                                             |
|---------------------|---------------------------------------------|
| Product Name        | Caelaris Linux (x86_64)                     |
| Version / Target    | v1.0-Rolling (Kernel 6.x)                   |
| Distribution Base   | Arch Linux Rolling Baseline                 |
| Document Author     | Caelaris Core Architecture Team             |
| Default Desktop     | KDE Plasma 6 (Wayland)                      |
| Alternative Desktop | GNOME 4x (Wayland)                          |
| Installer Subsystem | Caelaris Native Graphical Installer (PyQt6) |
| Date Generated      | September 2026                              |
| Status              | Approved for Release / Production           |

# 1. Executive Summary & Product Vision

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## 1.1 Vision Statement

**Caelaris Linux** is engineered to eliminate the traditional tradeoff between bleeding-edge software access and day-one usability. While Arch Linux offers unparalleled customization and up-to-date packages, its manual setup and configuration barrier prevent mainstream adoption. Turnkey distributions often introduce unwanted bloat, unoptimized kernels, or rigid desktop lock-in.

Caelaris delivers an ultra-responsive, modern, and gaming-optimized computing platform. It features out-of-the-box dual graphical sessions (KDE Plasma 6 as default and GNOME), instant session hot-switching, deep virtualization guest acceleration (VMware, VirtualBox, UTM/QEMU), complete GPU driver stacks (AMD, Intel, NVIDIA with DRM modesetting), custom PyQt6 graphical installer, and pro-grade performance tuning without bloat.

## 1.2 Core Value Proposition

- **Instant Gratification:** Zero-setup live preview with full 1080p dynamic auto-resizing, pre-configured live user, and full sudo privilege.
- **Freedom of Desktop Choice:** Boot into either KDE Plasma 6 or GNOME from the bootloader, switch seamlessly during preview via desktop shortcut, and choose desktop environment from SDDM login screen after install.
- **Custom Graphical Installer:** Native Python/PyQt6 installer ( `caelaris-installer-gui` ) supporting guided Btrfs subvolumes, ext4, EFI/BIOS bootloader generation, hardware driver activation, and single user profile cleanup.
- **Pro-Grade Gaming Performance:** Pre-tuned kernel sysctl parameters (huge map counts, low swappiness, Google BBR congestion, CAKE packet scheduling), PipeWire low-latency audio, GameMode, MangoHud, Steam, and VKD3D.
- **Universal Hardware & Hypervisor Integration:** Automated hardware driver detection for physical PCs alongside guest integration suites (VMware `open-vm-tools` , VirtualBox guest services, UTM/SPICE agents).
- **Clean Silent Direct Boot:** Streamlined GRUB bootloader displaying strictly `Caelaris Linux` , transitioning silently into the login screen without scrolling text or loading clutter.

## 2. Target Personas & Success Metrics

### 2.1 Target User Personas

| User Persona                       | Key Needs & Pain Points                                                                                                                            | How Caelaris Addresses It                                                                                                                                                                                   |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The Linux Gamer                    | Needs maximum FPS, smooth frame times, no shader compilation stutter, working Steam/Proton/Wine, and easy performance monitoring.                  | Pre-configured <code>vm.max_map_count=2147483642</code> , Steam, MangoHud, GameMode, 32-bit Vulkan drivers, and low-latency audio stack out-of-the-box.                                                     |
| The Power User / Dev               | Demands latest packages, fast compilation, AUR software access, modern CLI tooling, and responsive terminal environments.                          | Integrated <code>yay</code> AUR helper, modern Zsh/Bash with auto-completions, Fastfetch, btop, and Git.                                                                                                    |
| Virtualization & Evaluation Tester | Evaluates Linux inside VMware Workstation, VirtualBox, or UTM (macOS Apple Silicon/Intel) and hates manual resolution fixes or clipboard breakage. | Pre-activated <code>open-vm-tools</code> , <code>vboxservice</code> , <code>spice-vdagent</code> , and <code>caelaris-autoresize</code> ensuring immediate 1920x1080 resolution and two-way clipboard sync. |

### 2.2 Key Performance Indicators (KPIs) & Success Criteria

| Metric                        | Target SLA             | Implementation Strategy                                                                                                                       |
|-------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Cold Boot to Desktop (Live)   | < 5 seconds (NVMe SSD) | Elimination of Plymouth graphical locks, pre-created <code>liveuser</code> via systemd-sysusers, parallelized asynchronous systemd services.  |
| Display Auto-Resize Latency   | < 1 second             | Dedicated userspace agent ( <code>caelaris-autoresize</code> ) detecting VMware and VirtualBox monitor geometry changes.                      |
| Installer Completion Rate     | > 99% Success          | Custom PyQt6 installer with robust unsquashfs extraction, verified UUID bootloader injection, and driver configuration.                       |
| Installed User Profile Purity | 100% Single User       | Live user accounts completely purged from AccountsService, sysusers, and shadow database during install Step 6.                               |
| Bootloader Simplicity         | 1 Clean Title          | Configured as <code>Caelaris Linux</code> with silent boot flags ( <code>quiet loglevel=3 rd.udev.log_level=3 systemd.show_status=0</code> ). |

### 3. System Architecture & Technical Specifications

#### 3.1 Architecture Overview

Caelaris Linux utilizes a modular build architecture layered on top of the official Arch Linux `mkarchiso` foundation. All core operating system packages, performance configurations, desktop profiles, and installer logic are separated into isolated layers:

```
caelaris-linux/
├── shared/
│   ├── packages.common      # Unified base, gaming, media & hardware stack
│   ├── pacman.conf          # Multi-mirror package repos + Multilib enabled
│   ├── branding/            # Distro identity, os-release, logos
│   └── airootfs/             # Overlay root filesystem
│       ├── etc/default/grub  # Silent boot parameters & GRUB styling
│       ├── etc/sysctl.d/     # Pro-grade low-latency gaming & network sysctl
│       ├── etc/sysusers.d/   # Native systemd liveuser provisioning
│       └── usr/bin/           # Caelaris utilities (installer-gui, welcome, autoresize)
├── profiles/
│   ├── kde/                  # Primary ISO profile (KDE Plasma 6 + GNOME dual session)
│   │   ├── packages.x86_64   # Desktop-specific environment packages
│   │   └── profiledef.sh      # Zstd-19 compression & file permissions
│   └── .github/workflows/
│       └── build.yml          # CI/CD build runner & chunked release distributor
```

#### 3.2 Core Subsystem Specifications

| Subsystem              | Included Components                                                                                                                        | Key Capabilities                                                                                                   |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Base Kernel & System   | Linux Kernel 6.x, Linux-Firmware, SOF Audio Firmware, AMD/Intel Microcode, Btrfs-progs, e2fsprogs, exFAT, NTFS-3G.                         | Hardware auto-detection, universal filesystem support, CPU microcode security patching on boot.                    |
| Graphics & Display     | Mesa (Gallium), Vulkan-Radeon, Vulkan-Intel, Intel-Media-Driver, NVIDIA-Utills, lib32-NVIDIA-Utills, Wayland, XWayland, xf86-video-vmware. | Native Wayland compositing, zero-tear gaming, Vulkan compute, hardware video decode acceleration (VA-API / VDPAU). |
| Virtualization Support | <code>open-vm-tools</code> , <code>gtkmm3</code> , <code>vboxservice</code> , <code>spice-vdagent</code> , <code>qemu-guest-agent</code> . | Seamless mouse integration, dynamic resolution matching host window, shared folders, host-guest clipboard.         |
| Gaming Ecosystem       | Steam, GameMode, MangoHud (64 & 32-bit), VKD3D (DirectX 12 to Vulkan), Gamescope micro-compositor, Fuse2.                                  | Full Proton / Steam Play capability, per-game process scheduling, in-game FPS and hardware telemetry overlay.      |
| Audio Architecture     | PipeWire, WirePlumber, PipeWire-Pulse, PipeWire-ALSA, PipeWire-JACK.                                                                       | Ultra-low latency audio graph, professional JACK routing, seamless Bluetooth audio switching.                      |
| Application Management | Pacman, <code>yay</code> (native AUR helper), Flathub App Store backend.                                                                   | Access to over 100,000+ Linux packages via official Arch repositories, Flathub sandbox, and Arch User Repository.  |

## 4. Detailed Functional Requirements

### 4.1 Dual Desktop Environments & Live Session Switcher (FR-1)

**Requirement FR-1:** The distribution must support both KDE Plasma 6 and GNOME sessions within a single unified ISO, both in live preview and post-installation.

- **Bootloader Menu Selection:** Both systemd-boot (UEFI) and Syslinux (BIOS) bootloader menus present distinct options:
  - `Caelaris Linux (KDE Plasma 6 - Default)` : Loads Plasma 6 on Wayland via SDDM.
  - `Caelaris Linux (GNOME Desktop)` : Loads GNOME Shell on Wayland via SDDM.
  - `Caelaris Linux (Copy to RAM)` : Loads the entire system into RAM for portable high-speed usage.
- **1-Click Desktop Switcher ( `caelaris-switch-desktop` ):** Users can switch between KDE and GNOME during live preview without rebooting. Launching the desktop shortcut reconfigures SDDM autologin and restarts the display manager in 2 seconds.
- **Post-Install Dual Session:** The installed SDDM greeter presents an interactive session dropdown, allowing users to choose between Plasma Wayland and GNOME Wayland at every login.

### 4.2 Caelaris Graphical Installer & Profile Cleanup (FR-2)

**Requirement FR-2:** Python/PyQt6 graphical installer capable of partition automation, kernel deployment, driver enablement, and single-user profile enforcement.

- **Automated Guided Partitioning:** Creates EFI boot partition and root partition with choice of Btrfs subvolume layout or clean Ext4.
- **Single User Profile Guarantee:** Completely removes live user accounts ( `userdel -r -f liveuser` ) and deletes sysusers / AccountsService references so only the user-configured account appears at the SDDM login screen.
- **Clean Desktop Post-Install:** Removes all installer shortcuts ( `install-caelaris.desktop` ) from desktop and skeleton directories upon completion.
- **Driver Enablement:** Automatically enables hypervisor guest tools ( `vmtoolsd` , `vboxservice` , `spice-vdagentd` , `qemu-guest-agent` ) and configures NVIDIA DRM modeset if NVIDIA hardware is detected.

### 4.3 Silent Direct Boot & Single Bootloader Entry (FR-3)

**Requirement FR-3:** Bootloader must display strictly "Caelaris Linux" and boot cleanly into the graphical login screen without console clutter.

- **Streamlined Titles:** Generates GRUB entries cleaned of redundant strings, displaying only `Caelaris Linux` (or `Caelaris Linux (Recovery Mode)` for fallback).
- **Silent Kernel Parameters:** Appends `quiet loglevel=3 rd.udev.log_level=3 systemd.show_status=0 vt.global_cursor_default=0 video=1920x1080` for a seamless transition to SDDM.

- **Universal Removable Fallback:** Installs both NVRAM and `--removable` fallback bootloader ( `EFI/BOOT/BOOTX64.EFI` ) to prevent UEFI boot loops in VMware and VirtualBox.

## 4.4 Welcome Assistant One-Time Display (FR-4)

**Requirement FR-4:** The Caelaris Welcome Assistant must appear only once upon first login on installed systems.

- Tracks display state via `~/.config/caelaris-welcome-shown` .
- Exits immediately on subsequent logins in both KDE Plasma 6 and GNOME sessions.

# 5. Release Engineering & Continuous Integration

## 5.1 Multi-Part Release & 2GB Limit Bypass

**GitHub Releases Asset Limit:** GitHub enforces a 2,147,483,648 byte limit per asset. Caelaris Linux builds are divided automatically into <1.9GB chunks.

- **Chunk Generation:** Automated splitting via `split -b 1900M -d` produces `.part-00`, `.part-01`, and `.part-02`.
- **1-Click Reassembly:** Packages `combine.bat` for Windows users, using binary copy concatenation ( `copy /b *.part-* caelaris-kde.iso` ) and SHA256 checksum verification.
- **Direct Mirror:** Continuous release uploads to Pixeldrain API to provide a direct 1-click download of the complete 4.2GB ISO.

## 5.2 Project Roadmap & Milestones

| Milestone                 | Key Deliverables                                                                                                                           | Status          | Target  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------|
| Phase 1: Foundation       | Archiso base, KDE Plasma 6, pacman mirrors, base packages.                                                                                 | COMPLETED       | Q3 2026 |
| Phase 2: Dual DE          | GNOME 4x integration, live preview session hot-switcher.                                                                                   | COMPLETED       | Q3 2026 |
| Phase 3: Native Installer | PyQt6 GUI installer, silent boot, single profile cleanup.                                                                                  | COMPLETED       | Q3 2026 |
| Phase 4: Release Pipeline | GitHub Actions CI, chunked release assets, combine.bat.                                                                                    | COMPLETED       | Q3 2026 |
| Phase 5: Kernel & Theme   | Custom Zen/BORE scheduler kernel, bespoke Caelaris theme.                                                                                  | ROADMAP         | Q4 2026 |
| Phase 6: ARM64 & SBCs     | Caelaris Linux Raspberry Pi 5, 4 & 3B+ Edition (Arch Linux ARM, Broadcom VideoCore VII/VI/IV acceleration, zstd ZRAM, .img.xz & UEFI ISO). | ACTIVE PIPELINE | Q4 2026 |

**Conclusion:** Caelaris Linux establishes a modern, uncompromising standard for Arch-based Linux distributions, delivering gaming excellence, hypervisor agility, and desktop versatility in a polished, production-grade ISO.